basis vectors $\boldsymbol{\varphi}_1, \ldots, \boldsymbol{\varphi}_M$ a similar intuition holds, namely that if a particular basis vector is poorly aligned with the data vector $\mathbf{t}$, then it is likely to be pruned from the model.

We now investigate the mechanism for sparsity from a more mathematical perspective, for a general case involving M basis functions. To motivate this analysis we first note that, in the result (7.87) for re-estimating the parameter α_i, the terms on the right-hand side are themselves also functions of α_i. These results therefore represent implicit solutions, and iteration would be required even to determine a single α_i with all other α_j for $j \neq i$ fixed.

This suggests a different approach to solving the optimization problem for the RVM, in which we make explicit all of the dependence of the marginal likelihood (7.85) on a particular α_i and then determine its stationary points explicitly (Faul and Tipping, 2002; Tipping and Faul, 2003). To do this, we first pull out the contribution from α_i in the matrix $\mathbf{C}$ defined by (7.86) to give

$$\begin{aligned} \mathbf{C} &= \beta^{-1}\mathbf{I} + \sum_{j \neq i} \alpha_j^{-1} \boldsymbol{\varphi}_j \boldsymbol{\varphi}_j^{\mathrm{T}} + \alpha_i^{-1} \boldsymbol{\varphi}_i \boldsymbol{\varphi}_i^{\mathrm{T}} \\ &= \mathbf{C}_{-i} + \alpha_i^{-1} \boldsymbol{\varphi}_i \boldsymbol{\varphi}_i^{\mathrm{T}} \end{aligned} \tag{7.93}$$

where $\boldsymbol{\varphi}_i$ denotes the ith column of $\boldsymbol{\Phi}$, in other words the N-dimensional vector with elements $(\phi_i(\mathbf{x}_1), \ldots, \phi_i(\mathbf{x}_N))$, in contrast to $\boldsymbol{\phi}_n$, which denotes the nth row of $\boldsymbol{\Phi}$. The matrix $\mathbf{C}_{-i}$ represents the matrix $\mathbf{C}$ with the contribution from basis function i removed. Using the matrix identities (C.7) and (C.15), the determinant and inverse of $\mathbf{C}$ can then be written

$$|\mathbf{C}| = |\mathbf{C}_{-i}||1 + \alpha_i^{-1} \boldsymbol{\varphi}_i^{\mathrm{T}} \mathbf{C}_{-i}^{-1} \boldsymbol{\varphi}_i| \tag{7.94}$$

$$\mathbf{C}^{-1} = \mathbf{C}_{-i}^{-1} - \frac{\mathbf{C}_{-i}^{-1} \boldsymbol{\varphi}_i \boldsymbol{\varphi}_i^{\mathrm{T}} \mathbf{C}_{-i}^{-1}}{\alpha_i + \boldsymbol{\varphi}_i^{\mathrm{T}} \mathbf{C}_{-i}^{-1} \boldsymbol{\varphi}_i}. \tag{7.95}$$

Exercise 7.15

Using these results, we can then write the log marginal likelihood function (7.85) in the form

$$L(\boldsymbol{\alpha}) = L(\boldsymbol{\alpha}_{-i}) + \lambda(\alpha_i) \tag{7.96}$$

where $L(\boldsymbol{\alpha}_{-i})$ is simply the log marginal likelihood with basis function $\boldsymbol{\varphi}_i$ omitted, and the quantity $\lambda(\alpha_i)$ is defined by

$$\lambda(\alpha_i) = \frac{1}{2}\left[\ln \alpha_i - \ln(\alpha_i + s_i) + \frac{q_i^2}{\alpha_i + s_i}\right] \tag{7.97}$$

and contains all of the dependence on α_i. Here we have introduced the two quantities

$$s_i = \boldsymbol{\varphi}_i^{\mathrm{T}} \mathbf{C}_{-i}^{-1} \boldsymbol{\varphi}_i \tag{7.98}$$

$$q_i = \boldsymbol{\varphi}_i^{\mathrm{T}} \mathbf{C}_{-i}^{-1} \mathbf{t}. \tag{7.99}$$

Here s_i is called the *sparsity* and q_i is known as the *quality* of $\boldsymbol{\varphi}_i$, and as we shall see, a large value of s_i relative to the value of q_i means that the basis function $\boldsymbol{\varphi}_i$

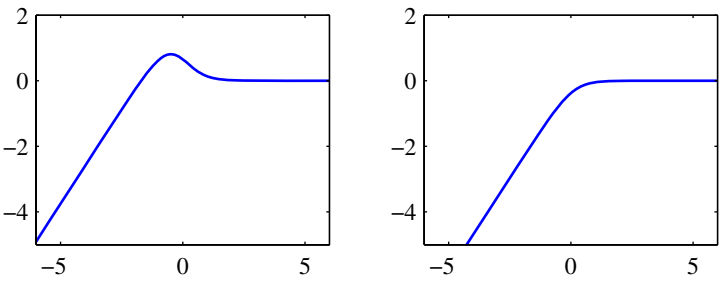


Figure 7.11 Plots of the log marginal likelihood $\lambda(\alpha_i)$ versus $\ln \alpha_i$ showing on the left, the single maximum at a finite α_i for $q_i^2 = 4$ and $s_i = 1$ (so that $q_i^2 > s_i$) and on the right, the maximum at $\alpha_i = \infty$ for $q_i^2 = 1$ and $s_i = 2$ (so that $q_i^2 < s_i$).

is more likely to be pruned from the model. The 'sparsity' measures the extent to which basis function $\boldsymbol{\varphi}_i$ overlaps with the other basis vectors in the model, and the 'quality' represents a measure of the alignment of the basis vector $\boldsymbol{\varphi}_n$ with the error between the training set values $\mathbf{t} = (t_1, \ldots, t_N)^{\mathrm{T}}$ and the vector $\mathbf{y}_{-i}$ of predictions that would result from the model with the vector $\boldsymbol{\varphi}_i$ excluded (Tipping and Faul, 2003).

The stationary points of the marginal likelihood with respect to α_i occur when the derivative

$$\frac{\mathrm{d}\lambda(\alpha_i)}{\mathrm{d}\alpha_i} = \frac{\alpha_i^{-1} s_i^2 - (q_i^2 - s_i)}{2(\alpha_i + s_i)^2} \tag{7.100}$$

is equal to zero. There are two possible forms for the solution. Recalling that $\alpha_i \geqslant 0$, we see that if $q_i^2 < s_i$, then $\alpha_i \to \infty$ provides a solution. Conversely, if $q_i^2 > s_i$, we can solve for α_i to obtain

$$\alpha_i = \frac{s_i^2}{q_i^2 - s_i}. \tag{7.101}$$

These two solutions are illustrated in Figure 7.11. We see that the relative size of the quality and sparsity terms determines whether a particular basis vector will be pruned from the model or not. A more complete analysis (Faul and Tipping, 2002), based on the second derivatives of the marginal likelihood, confirms these solutions are indeed the unique maxima of $\lambda(\alpha_i)$.

Exercise 7.16

Note that this approach has yielded a closed-form solution for α_i, for given values of the other hyperparameters. As well as providing insight into the origin of sparsity in the RVM, this analysis also leads to a practical algorithm for optimizing the hyperparameters that has significant speed advantages. This uses a fixed set of candidate basis vectors, and then cycles through them in turn to decide whether each vector should be included in the model or not. The resulting sequential sparse Bayesian learning algorithm is described below.

Sequential Sparse Bayesian Learning Algorithm

1. If solving a regression problem, initialize β.
2. Initialize using one basis function $\boldsymbol{\varphi}_1$, with hyperparameter α_1 set using (7.101), with the remaining hyperparameters α_j for $j \neq i$ initialized to infinity, so that only $\boldsymbol{\varphi}_1$ is included in the model.

3. Evaluate $\boldsymbol{\Sigma}$ and $\mathbf{m}$, along with q_i and s_i for all basis functions.
4. Select a candidate basis function $\boldsymbol{\varphi}_i$.
5. If $q_i^2 > s_i$, and $\alpha_i < \infty$, so that the basis vector $\boldsymbol{\varphi}_i$ is already included in the model, then update α_i using (7.101).
6. If $q_i^2 > s_i$, and $\alpha_i = \infty$, then add $\boldsymbol{\varphi}_i$ to the model, and evaluate hyperparameter α_i using (7.101).
7. If $q_i^2 \leqslant s_i$, and $\alpha_i < \infty$ then remove basis function $\boldsymbol{\varphi}_i$ from the model, and set $\alpha_i = \infty$.
8. If solving a regression problem, update β.
9. If converged terminate, otherwise go to 3.

Note that if $q_i^2 \leqslant s_i$ and $\alpha_i = \infty$, then the basis function $\boldsymbol{\varphi}_i$ is already excluded from the model and no action is required.

In practice, it is convenient to evaluate the quantities

$$Q_i = \boldsymbol{\varphi}_i^{\mathrm{T}} \mathbf{C}^{-1} \mathbf{t} \tag{7.102}$$

$$S_i = \boldsymbol{\varphi}_i^{\mathrm{T}} \mathbf{C}^{-1} \boldsymbol{\varphi}_i. \tag{7.103}$$

The quality and sparseness variables can then be expressed in the form

$$q_i = \frac{\alpha_i Q_i}{\alpha_i - S_i} \tag{7.104}$$

$$s_i = \frac{\alpha_i S_i}{\alpha_i - S_i}. \tag{7.105}$$

Exercise 7.17

Note that when $\alpha_i = \infty$, we have $q_i = Q_i$ and $s_i = S_i$. Using (C.7), we can write

$$Q_i = \beta \boldsymbol{\varphi}_i^{\mathrm{T}} \mathbf{t} - \beta^2 \boldsymbol{\varphi}_i^{\mathrm{T}} \boldsymbol{\Phi} \boldsymbol{\Sigma} \boldsymbol{\Phi}^{\mathrm{T}} \mathbf{t} \tag{7.106}$$

$$S_i = \beta \boldsymbol{\varphi}_i^{\mathrm{T}} \boldsymbol{\varphi}_i - \beta^2 \boldsymbol{\varphi}_i^{\mathrm{T}} \boldsymbol{\Phi} \boldsymbol{\Sigma} \boldsymbol{\Phi}^{\mathrm{T}} \boldsymbol{\varphi}_i \tag{7.107}$$

where $\boldsymbol{\Phi}$ and $\boldsymbol{\Sigma}$ involve only those basis vectors that correspond to finite hyperparameters α_i. At each stage the required computations therefore scale like $O(M^3)$, where M is the number of active basis vectors in the model and is typically much smaller than the number N of training patterns.

7.2.3 RVM for classification

We can extend the relevance vector machine framework to classification problems by applying the ARD prior over weights to a probabilistic linear classification model of the kind studied in Chapter 4. To start with, we consider two-class problems with a binary target variable $t \in \{0, 1\}$. The model now takes the form of a linear combination of basis functions transformed by a logistic sigmoid function

$$y(\mathbf{x}, \mathbf{w}) = \sigma\left(\mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x})\right) \tag{7.108}$$

where $\sigma(\cdot)$ is the logistic sigmoid function defined by (4.59). If we introduce a Gaussian prior over the weight vector $\mathbf{w}$, then we obtain the model that has been considered already in Chapter 4. The difference here is that in the RVM, this model uses the ARD prior (7.80) in which there is a separate precision hyperparameter associated with each weight parameter.

In contrast to the regression model, we can no longer integrate analytically over the parameter vector $\mathbf{w}$. Here we follow Tipping (2001) and use the Laplace approximation, which was applied to the closely related problem of Bayesian logistic regression in Section 4.5.1.

Section 4.4

We begin by initializing the hyperparameter vector $\boldsymbol{\alpha}$. For this given value of $\boldsymbol{\alpha}$, we then build a Gaussian approximation to the posterior distribution and thereby obtain an approximation to the marginal likelihood. Maximization of this approximate marginal likelihood then leads to a re-estimated value for $\boldsymbol{\alpha}$, and the process is repeated until convergence.

Let us consider the Laplace approximation for this model in more detail. For a fixed value of $\boldsymbol{\alpha}$, the mode of the posterior distribution over $\mathbf{w}$ is obtained by maximizing

$$\begin{aligned}\ln p(\mathbf{w}|\mathbf{t},\boldsymbol{\alpha}) &= \ln\{p(\mathbf{t}|\mathbf{w})p(\mathbf{w}|\boldsymbol{\alpha})\} - \ln p(\mathbf{t}|\boldsymbol{\alpha}) \\ &= \sum_{n=1}^{N}\{t_n \ln y_n + (1-t_n)\ln(1-y_n)\} - \frac{1}{2}\mathbf{w}^{\mathrm{T}}\mathbf{A}\mathbf{w} + \text{const} \qquad (7.109)\end{aligned}$$

where $\mathbf{A} = \mathrm{diag}(\alpha_i)$. This can be done using iterative reweighted least squares (IRLS) as discussed in Section 4.3.3. For this, we need the gradient vector and Hessian matrix of the log posterior distribution, which from (7.109) are given by

Exercise 7.18

$$\nabla \ln p(\mathbf{w}|\mathbf{t},\boldsymbol{\alpha}) = \boldsymbol{\Phi}^{\mathrm{T}}(\mathbf{t}-\mathbf{y}) - \mathbf{A}\mathbf{w} \qquad (7.110)$$

$$\nabla\nabla \ln p(\mathbf{w}|\mathbf{t},\boldsymbol{\alpha}) = -\left(\boldsymbol{\Phi}^{\mathrm{T}}\mathbf{B}\boldsymbol{\Phi} + \mathbf{A}\right) \qquad (7.111)$$

where $\mathbf{B}$ is an $N \times N$ diagonal matrix with elements $b_n = y_n(1-y_n)$, the vector $\mathbf{y} = (y_1,\ldots,y_N)^{\mathrm{T}}$, and $\boldsymbol{\Phi}$ is the design matrix with elements $\Phi_{ni} = \phi_i(\mathbf{x}_n)$. Here we have used the property (4.88) for the derivative of the logistic sigmoid function. At convergence of the IRLS algorithm, the negative Hessian represents the inverse covariance matrix for the Gaussian approximation to the posterior distribution.

The mode of the resulting approximation to the posterior distribution, corresponding to the mean of the Gaussian approximation, is obtained setting (7.110) to zero, giving the mean and covariance of the Laplace approximation in the form

$$\mathbf{w}^{\star} = \mathbf{A}^{-1}\boldsymbol{\Phi}^{\mathrm{T}}(\mathbf{t}-\mathbf{y}) \qquad (7.112)$$

$$\boldsymbol{\Sigma} = \left(\boldsymbol{\Phi}^{\mathrm{T}}\mathbf{B}\boldsymbol{\Phi} + \mathbf{A}\right)^{-1}. \qquad (7.113)$$

We can now use this Laplace approximation to evaluate the marginal likelihood. Using the general result (4.135) for an integral evaluated using the Laplace approxi-

mation, we have

$$p(\mathbf{t}|\boldsymbol{\alpha}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w}|\boldsymbol{\alpha})\,\mathrm{d}\mathbf{w} \simeq p(\mathbf{t}|\mathbf{w}^\star)p(\mathbf{w}^\star|\boldsymbol{\alpha})(2\pi)^{M/2}|\boldsymbol{\Sigma}|^{1/2}. \tag{7.114}$$

Exercise 7.19

If we substitute for $p(\mathbf{t}|\mathbf{w}^\star)$ and $p(\mathbf{w}^\star|\boldsymbol{\alpha})$ and then set the derivative of the marginal likelihood with respect to α_i equal to zero, we obtain

$$-\frac{1}{2}(w_i^\star)^2 + \frac{1}{2\alpha_i} - \frac{1}{2}\Sigma_{ii} = 0. \tag{7.115}$$

Defining $\gamma_i = 1 - \alpha_i\Sigma_{ii}$ and rearranging then gives

$$\alpha_i^{\text{new}} = \frac{\gamma_i}{(w_i^\star)^2} \tag{7.116}$$

which is identical to the re-estimation formula (7.87) obtained for the regression RVM.

If we define

$$\widehat{\mathbf{t}} = \boldsymbol{\Phi}\mathbf{w}^\star + \mathbf{B}^{-1}(\mathbf{t} - \mathbf{y}) \tag{7.117}$$

we can write the approximate log marginal likelihood in the form

$$\ln p(\mathbf{t}|\boldsymbol{\alpha}, \beta) = -\frac{1}{2}\left\{N\ln(2\pi) + \ln|\mathbf{C}| + (\widehat{\mathbf{t}})^{\mathrm{T}}\mathbf{C}^{-1}\widehat{\mathbf{t}}\right\} \tag{7.118}$$

where

$$\mathbf{C} = \mathbf{B} + \boldsymbol{\Phi}\mathbf{A}\boldsymbol{\Phi}^{\mathrm{T}}. \tag{7.119}$$

This takes the same form as (7.85) in the regression case, and so we can apply the same analysis of sparsity and obtain the same fast learning algorithm in which we fully optimize a single hyperparameter α_i at each step.

Appendix A

Figure 7.12 shows the relevance vector machine applied to a synthetic classification data set. We see that the relevance vectors tend not to lie in the region of the decision boundary, in contrast to the support vector machine. This is consistent with our earlier discussion of sparsity in the RVM, because a basis function $\phi_i(\mathbf{x})$ centred on a data point near the boundary will have a vector $\boldsymbol{\varphi}_i$ that is poorly aligned with the training data vector $\mathbf{t}$.

One of the potential advantages of the relevance vector machine compared with the SVM is that it makes probabilistic predictions. For example, this allows the RVM to be used to help construct an emission density in a nonlinear extension of the linear

Section 13.3

dynamical system for tracking faces in video sequences (Williams *et al.*, 2005).

So far, we have considered the RVM for binary classification problems. For $K > 2$ classes, we again make use of the probabilistic approach in Section 4.3.4 in which there are K linear models of the form

$$a_k = \mathbf{w}_k^{\mathrm{T}}\mathbf{x} \tag{7.120}$$

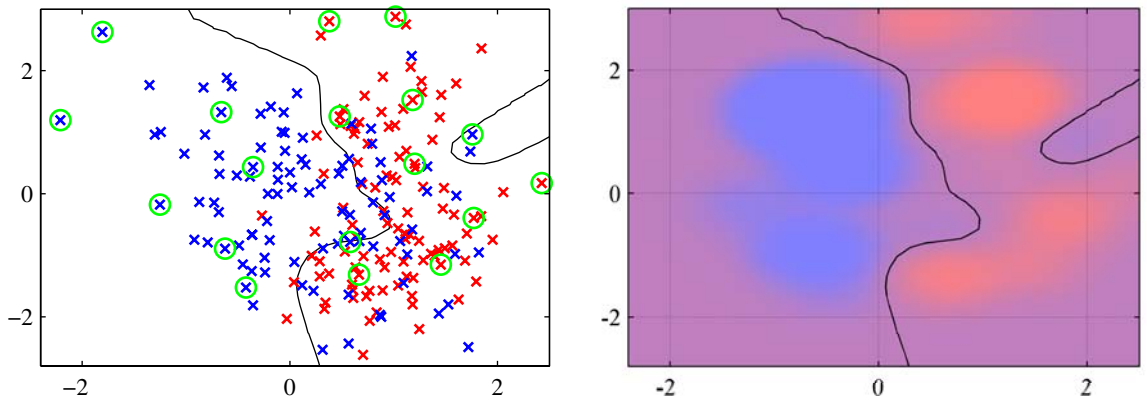


Figure 7.12 Example of the relevance vector machine applied to a synthetic data set, in which the left-hand plot shows the decision boundary and the data points, with the relevance vectors indicated by circles. Comparison with the results shown in Figure 7.4 for the corresponding support vector machine shows that the RVM gives a much sparser model. The right-hand plot shows the posterior probability given by the RVM output in which the proportion of red (blue) ink indicates the probability of that point belonging to the red (blue) class.

which are combined using a softmax function to give outputs

$$y_k(\mathbf{x}) = \frac{\exp(a_k)}{\sum_j \exp(a_j)}. \tag{7.121}$$

The log likelihood function is then given by

$$\ln p(\mathbf{T}|\mathbf{w}_1, \ldots, \mathbf{w}_K) = \prod_{n=1}^{N} \prod_{k=1}^{K} y_{nk}^{t_{nk}} \tag{7.122}$$

where the target values t_{nk} have a 1-of-K coding for each data point n, and $\mathbf{T}$ is a matrix with elements t_{nk}. Again, the Laplace approximation can be used to optimize the hyperparameters (Tipping, 2001), in which the model and its Hessian are found using IRLS. This gives a more principled approach to multiclass classification than the pairwise method used in the support vector machine and also provides probabilistic predictions for new data points. The principal disadvantage is that the Hessian matrix has size $MK \times MK$, where M is the number of active basis functions, which gives an additional factor of K^3 in the computational cost of training compared with the two-class RVM.

The principal disadvantage of the relevance vector machine is the relatively long training times compared with the SVM. This is offset, however, by the avoidance of cross-validation runs to set the model complexity parameters. Furthermore, because it yields sparser models, the computation time on test points, which is usually the more important consideration in practice, is typically much less.

Exercises

7.1 (⋆⋆) **www** Suppose we have a data set of input vectors $\{\mathbf{x}_n\}$ with corresponding target values $t_n \in \{-1, 1\}$, and suppose that we model the density of input vectors within each class separately using a Parzen kernel density estimator (see Section 2.5.1) with a kernel $k(\mathbf{x}, \mathbf{x}')$. Write down the minimum misclassification-rate decision rule assuming the two classes have equal prior probability. Show also that, if the kernel is chosen to be $k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^{\mathrm{T}}\mathbf{x}'$, then the classification rule reduces to simply assigning a new input vector to the class having the closest mean. Finally, show that, if the kernel takes the form $k(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^{\mathrm{T}}\boldsymbol{\phi}(\mathbf{x}')$, that the classification is based on the closest mean in the feature space $\boldsymbol{\phi}(\mathbf{x})$.

7.2 (⋆) Show that, if the 1 on the right-hand side of the constraint (7.5) is replaced by some arbitrary constant $\gamma > 0$, the solution for the maximum margin hyperplane is unchanged.

7.3 (⋆⋆) Show that, irrespective of the dimensionality of the data space, a data set consisting of just two data points, one from each class, is sufficient to determine the location of the maximum-margin hyperplane.

7.4 (⋆⋆) **www** Show that the value ρ of the margin for the maximum-margin hyperplane is given by

$$\frac{1}{\rho^2} = \sum_{n=1}^{N} a_n \tag{7.123}$$

where $\{a_n\}$ are given by maximizing (7.10) subject to the constraints (7.11) and (7.12).

7.5 (⋆⋆) Show that the values of ρ and $\{a_n\}$ in the previous exercise also satisfy

$$\frac{1}{\rho^2} = 2\widetilde{L}(\mathbf{a}) \tag{7.124}$$

where $\widetilde{L}(\mathbf{a})$ is defined by (7.10). Similarly, show that

$$\frac{1}{\rho^2} = \|\mathbf{w}\|^2. \tag{7.125}$$

7.6 (⋆) Consider the logistic regression model with a target variable $t \in \{-1, 1\}$. If we define $p(t = 1|y) = \sigma(y)$ where $y(\mathbf{x})$ is given by (7.1), show that the negative log likelihood, with the addition of a quadratic regularization term, takes the form (7.47).

7.7 (⋆) Consider the Lagrangian (7.56) for the regression support vector machine. By setting the derivatives of the Lagrangian with respect to $\mathbf{w}$, b, ξ_n, and $\widehat{\xi}_n$ to zero and then back substituting to eliminate the corresponding variables, show that the dual Lagrangian is given by (7.61).

7.8 ($\star$) **www** For the regression support vector machine considered in Section 7.1.4, show that all training data points for which $\xi_n > 0$ will have $a_n = C$, and similarly all points for which $\widehat{\xi}_n > 0$ will have $\widehat{a}_n = C$.

7.9 ($\star$) Verify the results (7.82) and (7.83) for the mean and covariance of the posterior distribution over weights in the regression RVM.

7.10 ($\star\star$) **www** Derive the result (7.85) for the marginal likelihood function in the regression RVM, by performing the Gaussian integral over $\mathbf{w}$ in (7.84) using the technique of completing the square in the exponential.

7.11 ($\star\star$) Repeat the above exercise, but this time make use of the general result (2.115).

7.12 ($\star\star$) **www** Show that direct maximization of the log marginal likelihood (7.85) for the regression relevance vector machine leads to the re-estimation equations (7.87) and (7.88) where γ_i is defined by (7.89).

7.13 ($\star\star$) In the evidence framework for RVM regression, we obtained the re-estimation formulae (7.87) and (7.88) by maximizing the marginal likelihood given by (7.85). Extend this approach by inclusion of hyperpriors given by gamma distributions of the form (B.26) and obtain the corresponding re-estimation formulae for $\boldsymbol{\alpha}$ and β by maximizing the corresponding posterior probability $p(\mathbf{t}, \boldsymbol{\alpha}, \beta|\mathbf{X})$ with respect to $\boldsymbol{\alpha}$ and β.

7.14 ($\star\star$) Derive the result (7.90) for the predictive distribution in the relevance vector machine for regression. Show that the predictive variance is given by (7.91).

7.15 ($\star\star$) **www** Using the results (7.94) and (7.95), show that the marginal likelihood (7.85) can be written in the form (7.96), where $\lambda(\alpha_n)$ is defined by (7.97) and the sparsity and quality factors are defined by (7.98) and (7.99), respectively.

7.16 ($\star$) By taking the second derivative of the log marginal likelihood (7.97) for the regression RVM with respect to the hyperparameter α_i, show that the stationary point given by (7.101) is a maximum of the marginal likelihood.

7.17 ($\star\star$) Using (7.83) and (7.86), together with the matrix identity (C.7), show that the quantities S_n and Q_n defined by (7.102) and (7.103) can be written in the form (7.106) and (7.107).

7.18 ($\star$) **www** Show that the gradient vector and Hessian matrix of the log posterior distribution (7.109) for the classification relevance vector machine are given by (7.110) and (7.111).

7.19 ($\star\star$) Verify that maximization of the approximate log marginal likelihood function (7.114) for the classification relevance vector machine leads to the result (7.116) for re-estimation of the hyperparameters.